

Data Product Manager Nanodegree

Applying Data Science to Product Management

Final Project: Developing an MVP Launch Strategy for a Flying Taxi Service

Welcome to your first week at Flyber

Flyber

In this project, you will apply the skills acquired in this course to create the MVP launch strategy for the first flying car taxi service, Flyber, in one of the most congested cities in America -- New York City.

You are responsible for bringing the first flying car taxi service to market by analyzing data and building a product proposal.

You will need to use the SQL workspace provided in the Classroom, and [Tableau Public](#), in order to successfully complete the project.

You'll present your answers, findings, and insights in the Answer Slides found in this deck. Feel free to include any additional slides, if needed.

Section 1: Data Exploration

Back to the basics of product management, identify your customer and their pain points:

- What are taxis used for?
- What are the characteristics of the users that leverage them?
- What are existing pain points with taxis?
- What are the existing pain points with digital ride-sharing services?

Ground travel creates uncertainty

USES: commuting, airport transfers, appointments, nightlife, tourism, and transit-gap trips.

USERS: residents and visitors without cars, time-sensitive riders, and people needing door-to-door access.

TAXI PAINS: road congestion, uncertain arrival time, variable availability, fare opacity, and driver idle/deadhead time.

RIDESHARE PAINS: surge pricing, cancellations, pickup mismatch, driver pay volatility, curb congestion, and privacy concerns.

What user improvements do you hypothesize a flying taxi service would have over the existing state of taxis today?

What market improvements do you hypothesize a flying taxi service would have the existing taxi service industry & physical road infrastructure today?

Flyber's advantage is time certainty

USER HYPOTHESIS: bypassing road bottlenecks will reduce travel-time variance on short, slow trips and make airport/business-district transfers more predictable.

DRIVER/PILOT HYPOTHESIS: scheduled nodes and reservations can reduce curb searching, cancellations, and unpaid deadhead time.

MARKET HYPOTHESIS: shifting selected trips off roads may reduce localized curb and road demand, but could create noise, airspace, safety, and access externalities.

These are testable hypotheses, not promised outcomes.

Upload [this dataset](#) into Tableau Online.

Ensure the fields are parsed correctly; field headers are included in the first row of the CSV.

Let's begin exploration!

Acquire a high-level understanding of the granularity and scope of the dataset, to inform the basis for your analyses:

- How many records are in the dataset
- What does each record represent?
- What is the primary key?
- What date range is your dataset bound to?
- What are the geographical bounds of this dataset? Is it limited to Manhattan, or is Brooklyn, Queens, Staten Island, the Bronx, and New Jersey included? Where are most of the data points centralized at? Are there outliers?

1,048,468 unique trips define the market

GRAIN: one completed taxi trip with pickup/dropoff time and coordinates, passengers, duration, and distance.

PRIMARY KEY: id; 1,048,468 records and 1,048,468 unique ids.

TIME: pickups from Jan 1 through Jun 30, 2016; the latest dropoff is Jul 1, 2016.

SPACE: trips center on Manhattan and extend to outer borough/airport catchments. Raw outlier bounds reach 32.18-51.88 N and 79.82-61.34 W.

QUALITY: 1,031,441 records (98.4%) pass the documented operational filter; 99.9% of pickups and 99.6% of dropoffs map to an NYC NTA.

You notice that the dataset does not contain explicit data points out-of-the-box, we'll need to enrich the dataset with relevant fields:

- You notice that ride price is not included, but figure it could be derived. Based on information about New York taxi prices gleaned from the internet, create a calculated field called `price` using the `duration`, `distance`, and `passenger count` fields.
- You hypothesize your target users will be those who take a relatively longer time getting to a destination that is relatively close, due to heavy traffic conditions and/or limitations to physical road infrastructure. To be able to analyze where this is happening, you will need to create a calculated field called `duration-to-distance ratio`.

Let's understand the scope and distribution various dimensions within the dataset. Calculate the **average**, **median**, and the **first & second standard deviation of the mean** for the following measures:

- duration
- distance
- passenger counts
- duration-to-distance ratio
- price

A typical trip is 11 min over 2.1 mi

Operational-quality subset: $n = 1,031,441$ (98.4% of records)

Duration (min): mean 14.04 | median 11.12 | SD 10.85

1 SD: 3.19 to 24.89 | 2 SD: -7.67 to 35.71

Distance (mi): mean 3.48 | median 2.13 | SD 3.92

1 SD: -0.44 to 7.41 | 2 SD: -4.37 to 11.33

Passengers: mean 1.66 | median 1.00 | SD 1.31

1 SD: 0.35 to 2.98 | 2 SD: -0.96 to 4.29

Minutes/mile: mean 5.39 | median 4.66 | SD 3.33

1 SD: 2.06 to 8.71 | 2 SD: -1.26 to 12.04

Fare proxy (USD): mean 12.94 | median 9.83 | SD 9.70

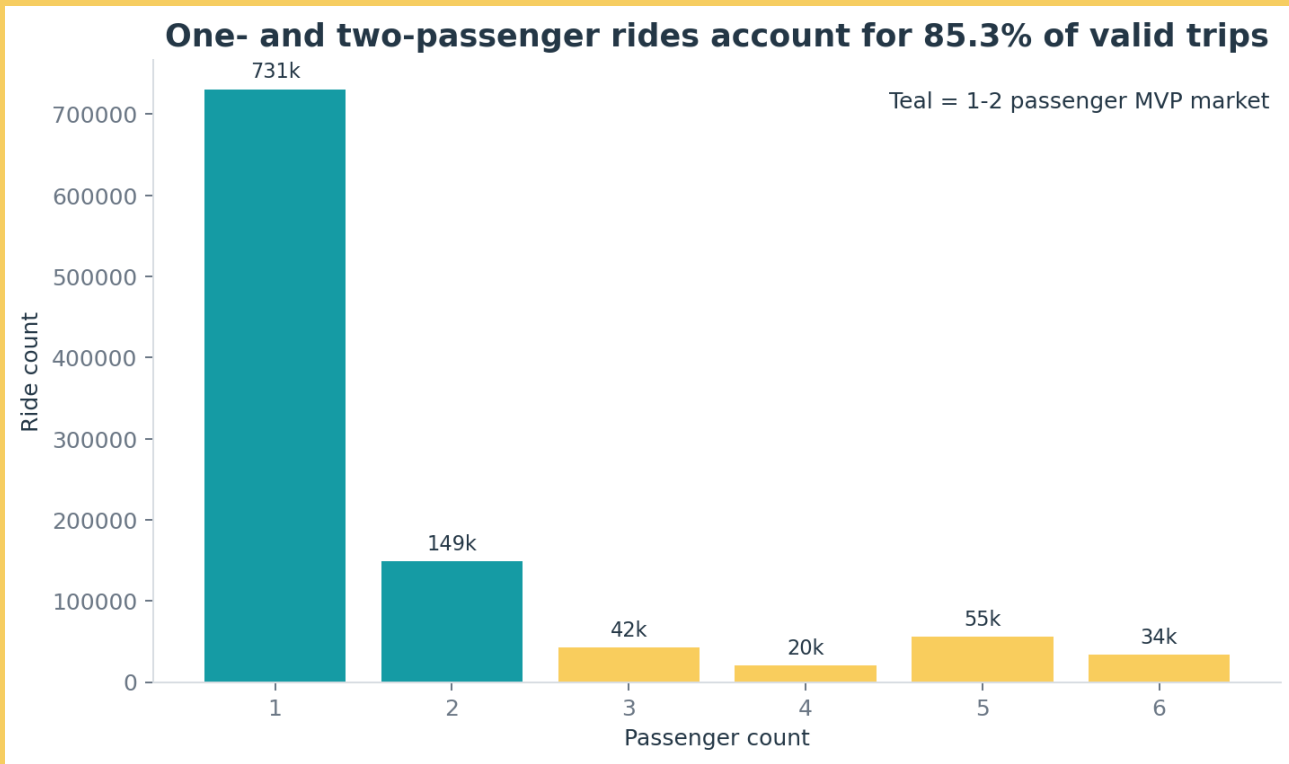
1 SD: 3.24 to 22.65 | 2 SD: -6.46 to 32.35

SD bands are descriptive and may cross impossible values; medians are retained for skewed measures.

Flying cars may have to have to be a lower weight for efficiency & take-off. Or you may just decide to leverage mini-copters for your initial MVP.

Create a histogram that visualizes the number of total rides grouped by passenger counts to analyze the potential market volume of low passenger pickups (1-2 passengers).

1-2 riders represent 85.3% of trips

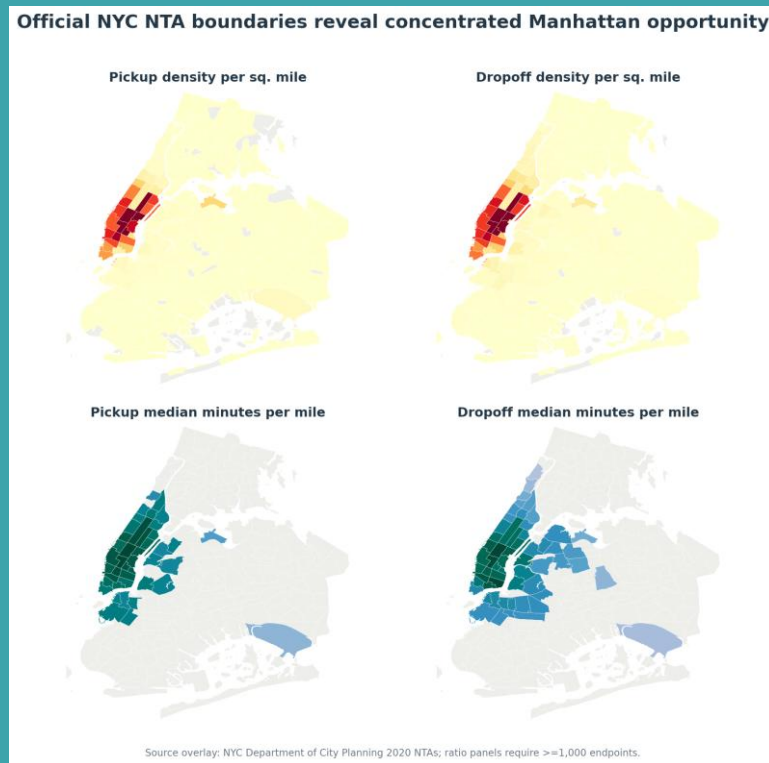


Full-data analysis; operational-quality subset $n = 1,031,441$.

For the initial MVP launch (& most likely GA), we have a finite amount of monetary resources to build Flyber pick-up / drop-off nodes. We'll need to be strategic on where we'll place them:

- Which neighborhoods/zip codes tends to experience a relatively higher density of pick-ups?
- Which neighborhoods/zip codes tends to experience a relatively higher density of drop-offs?
- Which neighborhoods/zip codes tends to have the highest duration-to-distance ratios, based on pick-up?
- Which neighborhoods/zip codes tends to have the highest duration-to-distance ratios, based on drop-off?
- For any of the neighborhoods identified, are there any potential areas within the neighborhood that are optimal for flying taxi pick-up / drop-off? What makes them suitable?

Midtown concentrates demand and delay



Midtown leads density and delay. [Open Tableau Public workbook.](#)

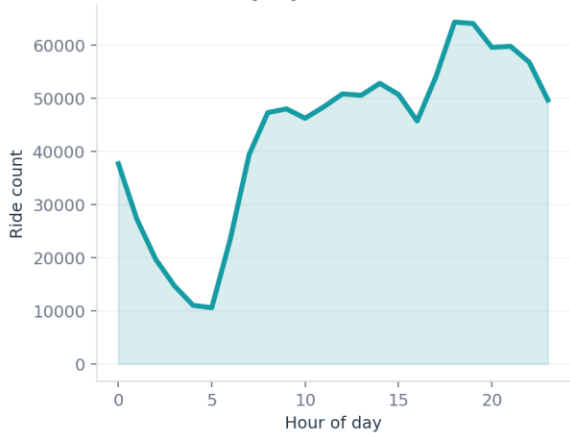
It may not make operational sense to have the service running 24/7, for now.

- What times throughout the day experience relatively higher volumes of ride pick-ups?
- What days throughout the week experience relatively higher volumes of ride pick-ups?
- Pinpoint any periods throughout the year that experience trend fluctuation or seasonality around ride pick-up volumes. This will help us in our post-launch analyses to determine if any spikes or dips were influenced by seasonality or through actual feature adoption/regression.

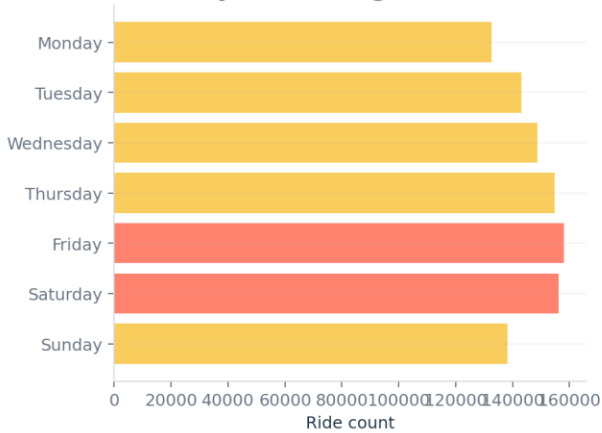
Thu-Sat evenings maximize demand

Temporal demand supports a Thursday-Saturday evening MVP

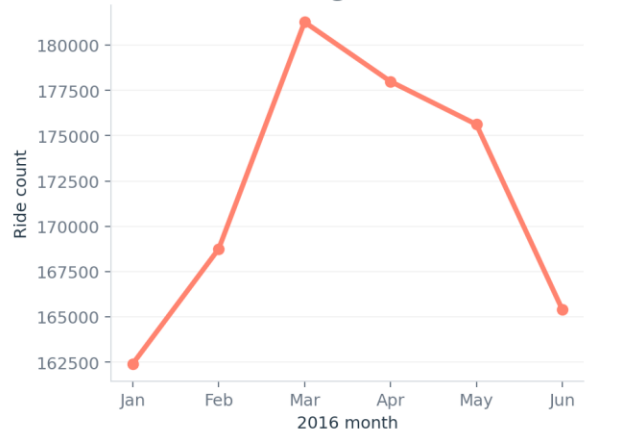
Pickups peak at 6-7 PM



Friday has the highest volume



Volume rises through March, then eases



6-7 PM leads hourly volume; Friday, Saturday, and Thursday lead days.

You and the user research team ran a quantitative survey on existing taxi and/or rideshare users in New York City to determine sentiment around potentially using a flying taxi service.

Dive into the survey results dataset in order to extract insights from explicit feedback.

Upload [this dataset](#) into Tableau Online or a SQL database (the classroom contains a workspace with the data for you as well).

Ensure the fields are parsed correctly, field headers are included in the first row of the CSV.

Question schema:

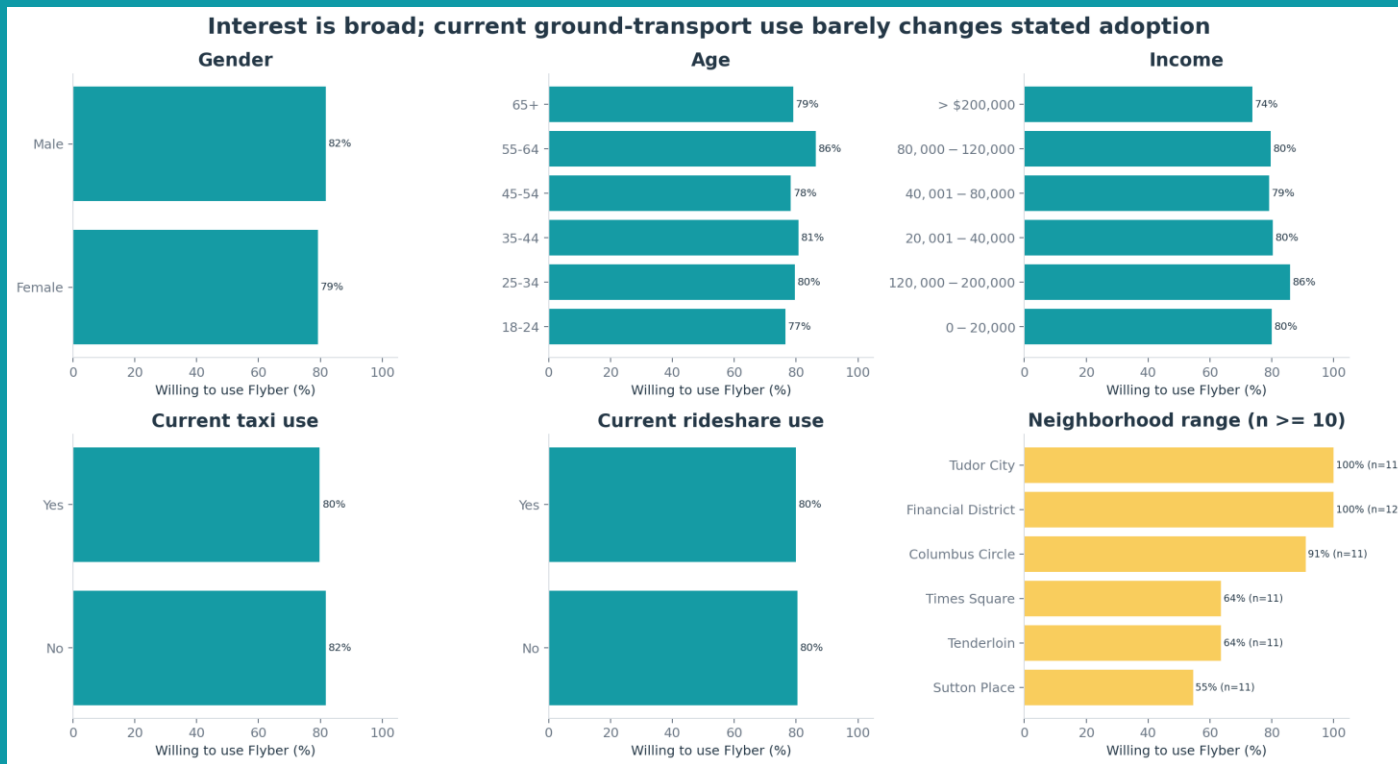
- Q1 - What is your email?
- Q2 - What gender do you identify as?
- Q3 - What is your age?
- Q4 - What is your annual income? (income bands)
- Q5 - What neighborhood do you reside in?
- Q6 - Do you currently use taxis? (Y/N)
- Q7 - Do you currently use ridesharing services? (Y/N)
- Q8 - Would you use a flying taxi service, if such a concept existed? (Y/N)
- Q9 - If yes to Q8, how much would you be willing to pay per mile for such a service? (USD)
- Q10 - If no to Q8, what is the reason?

To inform our future product marketing efforts, we'll want to extract the following:

- Is there an inclination of better Flyber adoption based on gender, age, income level, or neighborhood of residence?
- What is the distribution of potential price per mile based on gender, age, income level, and neighborhood of residence?
- What is the different personas/segments of negative sentiment towards not using a flying taxi car service?

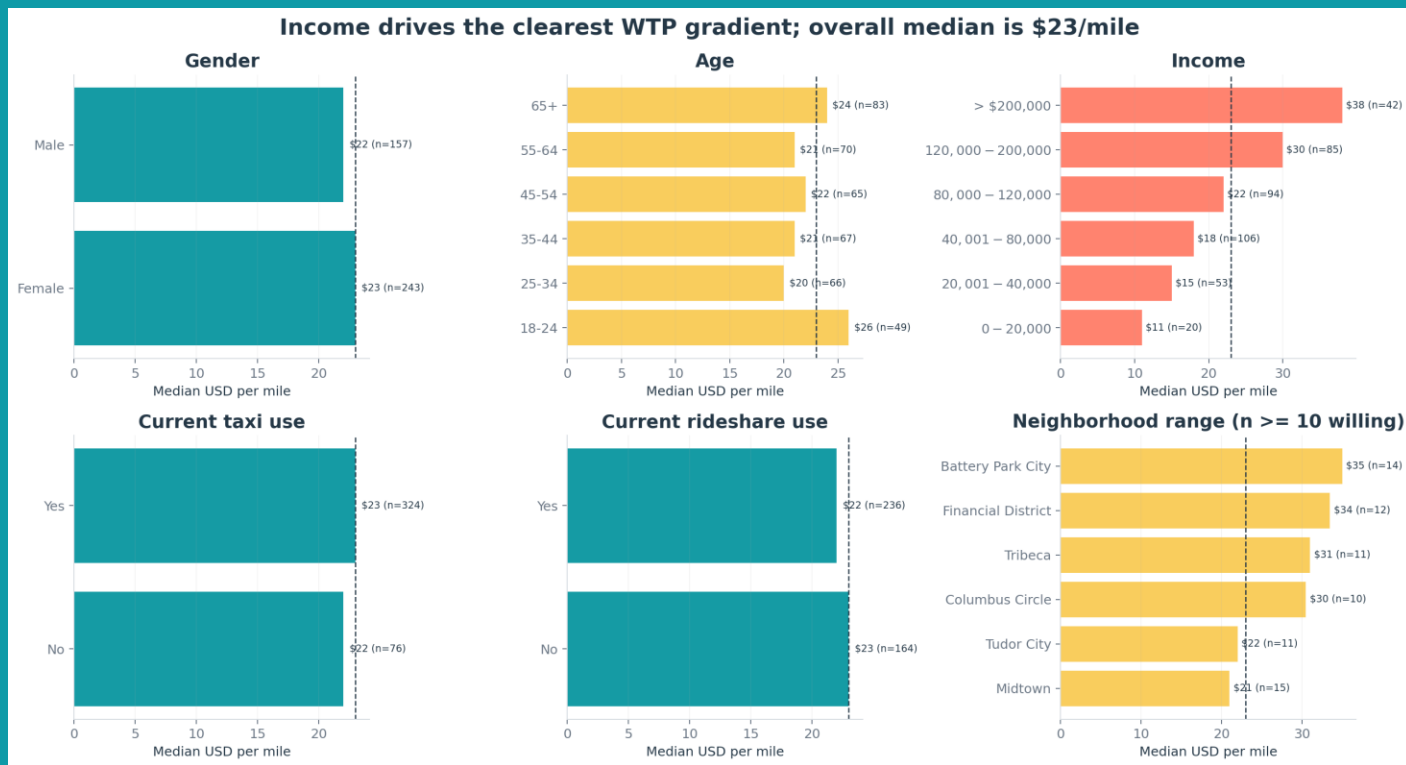
Please support your answers with clear, relevant visualizations.

Transport use barely changes interest



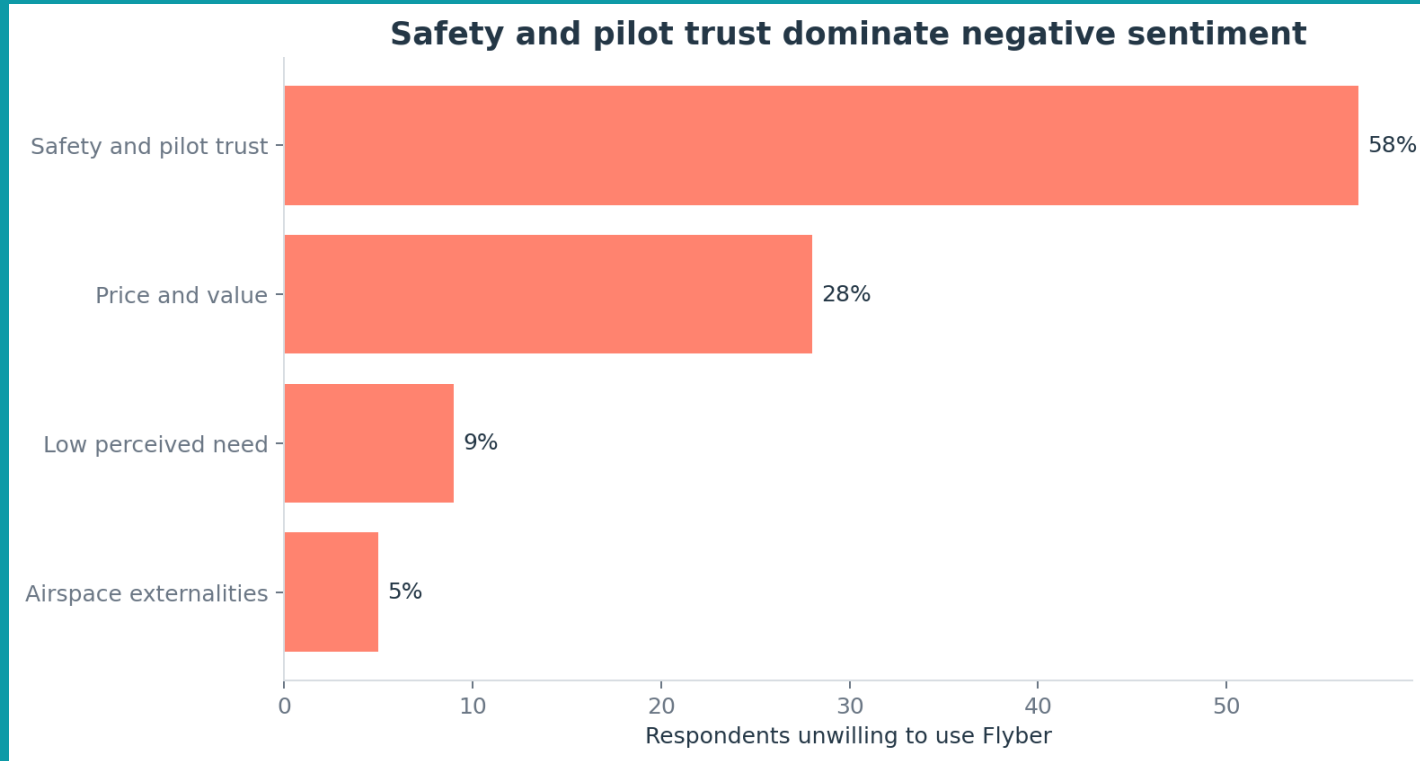
Taxi: 79.8% vs 81.7%; rideshare: 80.0% vs 80.4%. Neighborhoods: n >= 10.

Income predicts willingness to pay



\$23/mile overall; income leads differences; neighborhoods use n >= 10.

Safety drives 57.6% of objections



Negative-response segmentation: n = 99; safety and pilot trust lead.

Hooray! End of Section 1.

You will complete Section 2 at the end of this course.

Please submit this file for review for Section 1.